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Decoupling of subsoil carbon and nitrogen dynamics after long-term crop rotation and fertilization

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ABSTRACT

Enhancing global soil organic carbon storage by 4 per mille (‰) per annum would be enough to halt current net greenhouse gas emissions, but this goal seems lofty for conventional agriculture, which frequently results in soil organic carbon and nitrogen losses. Replacing mineral nitrogen with organic nitrogen sources may benefit soil carbon and nitrogen cycling in agricultural soils, but long-term effects are yet to be clearly demonstrated, especially in soils of high natural fertility. Here we report the effects of 34 years of legumes (crimson clover, fava beans) and non-legumes (maize) in rotation combined with different fertilization regimes (no fertilization, PK fertilization, NPK fertilization) on soil carbon and nitrogen storage throughout the uppermost meter of the soil profile. Fava beans did not enhance profile carbon storage. However, fava beans induced positive effects on subsoil nitrogen cycling, with lower subsoil nitrogen densities indicating lower nitrogen leaching potential. Incorporating a clover green mulch every 4 years enhanced organic carbon storage by an average of 4.1‰ per annum down the full meter of soil compared with a conventional maize rotation, but only combined with phosphorus and potassium fertilization. The enhancement was detected below the plough-horizon, indicating that merely sampling topsoil is insufficient to assess soil carbon dynamics in these arable soils. In contrast, maize contributed only a small portion to SOC, with subsoil C contributions negligible. These results indicate that a careful combination of long-term, site-adapted crop and fertilization management strategies can help enhance SOC storage in naturally fertile soils without apparent C deficit.

1. Introduction

Loss of soil organic carbon (SOC) is a threat to earth systems, as SOC fulfils a variety of ecosystem functions, including but not limited to boosting soil fertility, regulating climate and buffering soil systems against adverse climate impacts (Paustian et al., 2016). For decades, scientists across the globe have advocated enhancing SOC storage, and recently a global target of annual SOC enhancement of 4‰ has been calculated as being sufficient to offset current anthropogenic greenhouse gas emissions (Lima Paris Agreement Agenda, 2015; Lal, 2016).

One proposed strategy for climate-smart soil management is to reduce the application of mineral nitrogen fertilizer (N_{min}) to soils (Paustian et al., 2016). The conversion of atmospheric N_2 into N_{min} during the Haber-Bosch process was a key driver of the Green Revolution, which has fed a large portion of the world (Smil, 1999). However, this process is associated with high-energy consumption, accounting for at least 1% of global energy use. Around 80% of N_{min} produced in the Haber-Bosch process is applied to soils as fertilizer (Erisman et al., 2008) but, N_{min} use efficiency in agriculture is only around 33% (Raun & Johnson, 1999), with N use efficiency declining from ∼60% to ∼30% in the decades from the 1960s to 2000 (Erisman et al., 2008), indicating very high losses from soil. These N losses can have severe impacts on natural resources, as evident from losses of biodiversity, nitrate contamination of freshwater systems, emission of greenhouse gases to the atmosphere as well as soil acidification (Millenium Ecosystem Assessment, 2005). Furthermore, long-term N fertilization has been associated with changes to C cycling in soils (Neff et al., 2002) as well as losses of mineral-associated and subsoil organic carbon (Shahbaz et al., 2017). Supplementing or replacing agricultural N_{min} with alternative N sources, such as naturally converted atmospheric N or recycled waste N, may therefore help to minimize damage to the environment.

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A prime candidate for alternative N sources are legumes, which have been used to enhance and maintain agricultural productivity since ancient times as they naturally fix atmospheric N into organic biomass. A portion of this N is then released into the soil during turnover of soil organic matter (SOM). The ability of legumes to substitute for N fertilizers and therefore reduce associated energy consumption was illustrated by Peoples and Craswell (1992), who found that green manure legumes generally provide N equivalent to 50–120 kg ha^{-1}, but individual systems may receive over 500 kg ha^{-1} from just a few months of legume growth, with calculated N_{min} offsets equivalent to nearly 10,500 MJ ha^{-1} energy consumption (Liebman et al., 2012). More recently, the use of legumes in crop rotation has received increasing attention due to their at times deep rooting systems, that promote biopore formation and thus provide access to subsoil water and nutrient resources (Gaiser et al., 2012; Kautz et al., 2013).

Including legumes in rotation has also been found to reduce C and N losses from agroecosystems (Drinkwater et al., 1998), as well as enhance SOM storage (Gregorich et al., 2001; Robertson et al., 2015), as biomass turnover leads to the incorporation of OC into soil. Whilst legumes therefore have potential to boost soil fertility, results are not consistent and legumes do not always enhance C storage (Robertson et al., 2015). This inconsistency in SOC enhancement is likely due to other limiting factors to plant growth, such as climatic conditions or stoichiometric constraints on biomass production (nutrient limitation). P fertilization, in particular, has been positively associated with legume nodulation and yields (Wani et al., 1995). Enhanced K availability has also been shown to stimulate red clover growth (Oram et al., 2014). The latter study showed that growth stimulation was only achieved under N limitation, highlighting the complex interactions in nutrient availability which affect legume growth. These studies highlight the need for more investigations aimed at disentangling the complex interactions between leguminous crops, nutrient dynamics and SOC storage, in particular for subsoils (Kautz et al., 2013), which are generally defined as lying below the A-horizon, i.e. the plough layer in agricultural soils.

One issue with identifying changes in subsoil OC storage is that subsoil OC contents tend to be low, whereas subsoil heterogeneity is high (Hobley and Wilson, 2016). This combination of low subsoil OC content, which is at times near the analytical limit of determination, and high variance results in a need to analyse more samples to detect the same changes due to management effects as in topsoil OC storage. To overcome these difficulties detecting differences in OC storage in subsoils, isotope analyses can be used to complement traditional chemical analysis (Hobley et al., 2017; Robinson, 2001). Such analyses are more sensitive than bulk OC contents, thereby improving analytical power.

In this study we evaluated the interactive effects of leguminous (crimson clover as green mulch, fava beans) and non-leguminous (maize) crops in rotation in combination with various inorganic fertilization regimes (no fertilization, PK fertilization, NPK fertilization) to assess the potential for additional SOC and N storage down the profile to 1 m depth. We hypothesized that legumes increase SOC and N storage, with benefits amplified by additional P and K fertilization, but reduced by additional N fertilization. However, the benefits of legumes on soil properties will be depth differentiated between crops, as clover green mulch is ploughed into the surface soil, whereas fava beans are harvested, so that N is primarily added via roots.

2. Methods

2.1. Site and soils

Sampling was performed in April 2016 at the Biological Nitrogen Fixation Experiment, located at and managed by the Justus von Liebig University, Gießen, Germany (longitude: 50° 36′ 12‘ ‘N, latitude: 8° 39′ 16‘ ‘E). The site is at 158 m above sea level and experiences a temperate climate (Köppen-Geiger climate classification Cfb) with a mean annual temperature of 9.6 °C and a mean annual precipitation of 666 mm (long-term average from 1939 to 2017, Deutscher Wetterdienst, 2017).

Table 1
Fully factorial design of experiment with varied first year crop crossed with fertilization regime.

First year crop	Fertilisation regime		
Crimson clover	None	PK	NPK
Fava beans	PK	NPK	None
Maize	NPK	None	PK

The soil developed upon heavy textured, fluvial material and was classified as a eutric fluvic gleyic Cambisol (WRB 2015). The texture throughout the profile was determined using Micromeritics Sedigraph III Plus particle-size analyser on 28 samples chosen randomly over all treatments and depth intervals (see below). The texture was classified as silty clay, without a gradient from surface to subsoil (sand: 8 ± 4% mass, silt: 44 ± 3% mass, clay: 48 ± 5% mass). Rock content was negligible (median < 1% mass). The dominant mineralogical components were determined by X-ray diffraction using a Phillips PW1070 diffractometer, which identified quartz, numerous phyllosilicates and clay mineral components including illite and muscovite, chlorite, kaolinite and feldspar in all depths, with the relative abundance of the latter increasing with depth.

The experiment was established in 1982 as a fully factorial split plot design, with spatially randomized field replicates in four blocks. The first factor was a four-year crop rotation system, in which the first year crop was varied and included one of two legumes (crimson clover, *Trifolium incarnatum*, or fava beans, *Vicia faba*) or maize (*Zea mais*) (Table 1). The clover was used as a green mulch, with the entire biomass ploughed into the soil (plough depth ~30 cm), whereas the fava and maize were harvested, with the standing residues incorporated into the soil after harvest. The crops in years 2–4, were winter wheat (*Triticum aestivum* ssp. *aestivum*), winter rye (*Secale cerale*), and summer barley (*Hordeum vulgare*) respectively. The second factor was fertilization. Three fertilization regimes were employed, including a zero fertilization control (‘no fertilization’ or ‘none’), a P and K fertilization without additional N (‘PK’), and a full mineral fertilization comprising N, P and K (‘NPK’). The fertilized plots received 90 kg ha^{-1} yr^{-1} P as triple superphosphate and 120 kg ha^{-1} yr^{-1} K as KCl. N fertilization was optimized to the winter wheat crop, receiving a total of 180 kg ha^{-1} yr^{-1} as calcium ammonium nitrate (to prevent acidification) in 3 doses. In year 1 (i.e. varied crop), no N fertilization occurred.

Each plot was 42 m^2 and each of the nine treatments (Table 1) was replicated in four blocks, resulting in 36 plots. The plots were sampled during a winter wheat crop using a hydraulic steel corer with an inner diameter of 6 cm. Two cores were taken from each plot to a depth of one metre and cut into four depth increments: 0–30 cm, 30–50 cm, 50–70 cm, 70–100 cm, yielding a total of 144 samples. These depths corresponded to the sampling scheme of the German Soil Inventory for Agriculture (https://www.thuenen.de/en/ak/projects/agricultural-soil-inventory-bze-lw/). Each depth increment was weighed in the field for bulk density calculations.

2.2. Sample treatment

Around 15 g of field fresh sample was weighed and dried at 105 °C to determine gravimetric moisture content. The remaining sample was dried at room temperature to constant mass. Dry samples were disaggregated using a jaw crusher prior to sieving at 2 mm. A 2 g aliquot of fine sample (< 2 mm) was ground using a Teflon ball mill to pass a 150 µm mesh for C, N and isotopic analysis.

Table 2
Correlation matrix and probability of error of correlation for soil properties. The values to the upper right of the diagonal are the Pearson's product moment (R) between two variables. The italicized values to the lower left of the diagonal are the probability of error (p-values) of the correlation.

R p	log Depth	pH	Electrical conductivity	Bulk density	SOC density	N density	CN ratio	$\delta^{13}C$	$\delta^{15}N$	P_{cal} concentration	K concentration
log Depth [log(m)]		0.22	−0.78	0.57	−0.93	−0.90	−0.79	0.37	0.20	−0.53	−0.84
pH	*0.01*		−0.27	0.01	−0.21	−0.20	−0.14	0.06	0.21	−0.43	−0.33
Electrical conductivity [μS cm^{-1}]	*< 0.01*	*< 0.01*		−0.61	0.77	0.73	0.66	−0.34	−0.44	0.51	0.87
Bulk density [g cm^{-3}]	*< 0.01*	*1.00*	*< 0.01*		−0.53	−0.53	−0.42	0.46	0.36	−0.10	−0.51
SOC density [kg m^{-3}]	*< 0.01*	*0.02*	*< 0.01*	*< 0.01*		0.97	0.86	−0.35	−0.10	0.47	0.83
N density [kg m^{-3}]	*< 0.01*	*0.02*	*< 0.01*	*< 0.01*	*< 0.01*		0.73	−0.35	−0.04	0.35	0.76
CN	*< 0.01*	*0.10*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*		−0.25	−0.13	0.57	0.73
$\delta^{13}C$ [‰$_{VDPB}$]	*< 0.01*	*0.46*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*		0.31	−0.17	−0.30
$\delta^{15}N$ [‰$_{Natm}$]	*0.03*	*< 0.01*	*< 0.01*	*< 0.01*	*0.33*	*0.88*	*0.16*	*< 0.01*		−0.34	−0.41
P_{cal} [mg kg^{-1}]	*< 0.01*	*< 0.01*	*< 0.01*	*0.29*	*< 0.01*	*< 0.01*	*0.04*	*< 0.01*	*< 0.01*		0.67
K [mg kg^{-1}]	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	*< 0.01*	

2.3. Physical and chemical analyses

Bulk density was calculated from the quotient of the mass and volume for each depth increment, corrected for moisture content. Fine samples were analysed for pH and electrical conductivity in a 1:5 (m:m) soil to water solution with deionized water.

Total carbon and nitrogen contents were analysed using dry combustion in a EURO EA elemental analyser. Calcimetry with 4 M HCl confirmed the absence of any inorganic carbon, so that total carbon is equal to organic carbon. From this the carbon density (kg m^{-3}), i.e. the specific carbon storage in a given profile depth, was determined as the product of the whole soil bulk density (ρ, kg m^{-3}) adjusted for the rock mass content (RC, g g^{-1}) and C concentrations of the fine soil ($SOC_{fine\ fraction}$, g g^{-1}) in each depth increment (Hobley et al., 2013):

$$Carbon\ \ density = SOC_{fine\ fraction}\bullet\rho\bullet(1{-}RC)$$

Nitrogen density was calculated analogously. Changes in SOC and N densities reflect the net changes in both the C and N concentrations and soil bulk density, overcoming issues associated with the inverse relationship between soil organic matter content and bulk density. Total C and N stocks were calculated as the cumulative sum of the product of the densities and individual sampling depths.

Carbon and nitrogen stable isotopic composition was analysed using a Delta C isotope ratio mass spectroscope. $\delta^{13}C$ is reported relative to Vienna Pee Dee Belemnite and $\delta^{15}N$ is reported relative to atmospheric nitrogen.

Calcium acetate lactate-extractable phosphorus (P_{CAL}) and potassium (K_{CAL}), considered as a proxy for plant-available P and K in German soils, were measured according to Schüller (1969). Five g of soil were treated with 100 ml of a buffered extraction solution containing calcium lactate, calcium acetate and acetic acid (pH 3.7–4.1), shaken for 2 h and filtered over a paper filter. P contents in the extraction solution were measured photometrically using a Tecan Infinite M200 PKro plate reader after reaction with molybdenum blue solution and ascorbic acid (Murphy and Riley, 1962), whereas K contents were measured using an AnalytikJena NovAA 400 P atomic absorption spectrometer.

2.4. Data analysis

Crop and fertilization effects on soil physical and chemical properties down the profile were tested using multi-factor analysis of variance / linear models. Models were fitted using first-year crop type and fertilization regime as fixed factors and the mean of the sampling depths as a continuous variable and interactions between predictors were tested, with the errors specified within the experimental blocks:

$$Response \sim Crop_{year\ 1}\bullet Fertilization\bullet Depth + Error(Block)$$

Where no significant interactions between treatment and depth were found, non-interactive models were fit:

$$Response \sim Crop_{year\ 1}\bullet Fertilization + Depth + Error(Block)$$

Lastly, where no significant interactions between treatment and depth were found, non-interactive models were fit:

$$Response \sim Crop_{year\ 1} + Fertilization + Depth + Error(Block)$$

Model fit and conformity to model assumptions were assessed using residual plots and where necessary, variables were log-transformed prior to model fitting. Where significant relationships between response and predictors variables were found, differences between groups were investigated based on model coefficients and using the Tukey's post-hoc analysis or Games-Howell post-hoc, where variance was not homogenous, after de-trending for depth relationships via partial correlation.

Additionally, regression tree analysis was undertaken for both C and N densities to identify relationships with cropping systems, fertilization and nutrient availability at different profile depths, with tree depths limited to 5 splits, splits constrained by likelihood (p terminal node < 0.1) and terminal nodes constrained to a minimum of 3 data points. Surrogate splits were also investigated.

Unless otherwise specified, the term significant is used to describe results with a probability of error $0.01 < p \le 0.05$, highly significant is used to describe results with a probability of error $p \le 0.01$. Where different models indicated significant effects (e.g. with and without interactions), we report the results indicated by most models as significant or likely ($0.05 \le p < 0.1$). All statistical analyses were performed using R software. The strength of relationships based on absolute values of Pearson's correlation (R) is described as $0 < |R| \le 0.2$: negligible; $0.2 < |R| \le 0.4$: weak; $0.4 < |R| \le 0.6$: moderate; $0.6 < |R| \le 0.8$: strong; $0.8 < |R| \le 1.0$: very strong.

Table 3
Coefficients of variation of soil properties at individual sampling depths.

Sampling depth [cm]	pH	Electrical conductivity	Bulk density	SOC concentration	SOC density	N concentration	N density	$\delta^{13}C$	$\delta^{15}N$	P_{cal} concentration	P_{cal} density	K concentration	K density
0–30	0.02	0.23	0.04	0.10	0.10	0.08	0.07	0.02	0.06	0.16	0.17	0.25	0.25
30–50	0.01	0.13	0.05	0.09	0.11	0.08	0.08	0.03	0.06	0.65	0.66	0.45	0.40
50–70	0.01	0.16	0.07	0.18	0.16	0.17	0.15	0.01	0.05	0.68	0.70	0.17	0.20
70–100	0.02	0.18	0.09	0.22	0.20	0.18	0.16	0.01	0.05	0.60	0.63	0.28	0.31
All depths	0.02	0.42	0.08	0.40	0.35	0.31	0.26	0.02	0.07	0.58	0.58	0.75	0.69

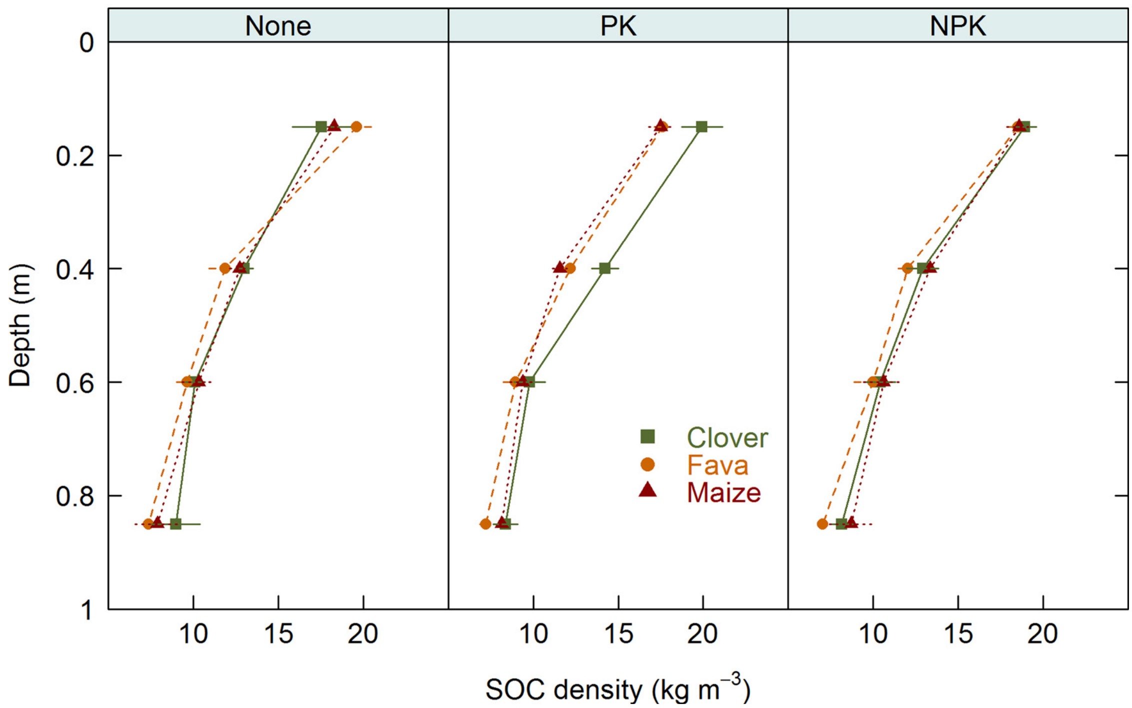

Fig. 1. Soil organic carbon density under different crop rotations systems and fertilization regimes. Dots represent the mean, error bars 1 sd.

3. Results

3.1. Depth distribution of soil physical and chemical properties

The majority of the investigated soil properties showed strong depth gradients (Table 2), although relationships were non-linear, with the best depth correlations found after log transformation of soil depth. The relationship between soil depth and bulk density was highly significant and positive, though only of moderate strength (Table 2). Highly significant, negative relationships between depth and numerous soil properties were found, including electrical conductivity, SOC density, N density, the CN-ratio, plant available P concentration and K concentration. In particular, the relationship between depth and SOC and N was very strong. In contrast to the relatively strong depth relationships outlined above, the depth trends of pH and both C and N stable isotopes were less clear. Though highly significant, $\delta^{13}C$ was only weakly positively associated with depth. Similarly, the relationships between both pH and the isotopic N-composition with depth were weak to negligible.

In general, the relative variance of all physical and chemical soil properties increased down the soil profile (Table 3), though for pH, and stable C and N isotopes the coefficients of variation were very low and not related with depth. Notably, below the plough horizon, the variance in the C and N densities was lower than for C and N concentrations.

3.2. Fertilization and crop rotation effects on pH, electrical conductivity and bulk density

Although weak, the linear models indicated a highly significant relationship between depth and pH. pH was highly significantly reduced by NPK fertilization compared with plots receiving no N fertilizer, although overall shifts were small (< 0.1 compared with both unfertilized and PK fertilized plots, Fig. S1). NPK fertilization was particularly acidifying within the maize system, where profile pH was reduced by up to 0.16 units compared with the fava and clover plots. Similarly, electrical conductivity was highly significantly increased by NPK fertilization down the profile, with an average increase of 9 ± 5 µS cm^{-1} compared with unfertilized plots and 7 ± 5 µS cm^{-1} compared with PK fertilized plots (Fig. S2). Electrical conductivity was not significantly affected by cropping system.

Bulk density was statistically indistinct between treatments at the surface, and did not significantly differ between cropping systems in the unfertilized plots (Fig. S3). However, in the subsoil of the fertilized plots, bulk densities were higher in the clover rotation compared with the fava and maize systems. Under PK fertilization, the subsoil bulk densities of the clover system were significantly greater than the maize system (0.11 ± 0.04 g cm^{-3}) and likely ($p = 0.06$) greater than the fava system (0.08 ± 0.04 g cm^{-3}). However, in the NPK fertilized plots, both the maize and the clover systems had increased subsoil bulk densities compared with the fava system, where the post-hoc analysis indicated a significant increase in bulk densities averaging

Table 4
C and N stocks [kg m^{-3}] to 1 m sampling depth under different treatments. Values are mean ± standard deviation of four replicates.

Fertilization	No fertilizer			PK fertilizer			NPK fertilizer		
Precrop	Clover	Fava	Maize	Clover	Fava	Maize	Clover	Fava	Maize
C stock	126 ± 11	126 ± 12	125 ± 14	133 ± 14	117 ± 8	119 ± 6	128 ± 11	121 ± 11	130 ± 17
N stock	15.0 ± 1.2	14.5 ± 0.6	15.1 ± 1.6	15.4 ± 1.1	13.6 ± 0.3	14.5 ± 0.6	15.4 ± 1.0	14.4 ± 1.0	15.6 ± 1.8

0.09 ± 0.02 gm cm-^{3} in the subsoil of the clover and maize plots compared with the fava plots. Despite the subsoil bulk density effect, cumulative soil mass to 1 m was not significantly affected by treatment.

3.3. Fertilization and crop rotation effects on soil organic carbon and soil nitrogen

Significant differences in C storage were found between treatments and depths. Under PK fertilization, C densities were significantly higher in the upper half of the soil profile within the clover system than either the fava or maize systems (Fig. 1, S4). Under PK fertilization the clover system stored 16 Mg C ha^{-1} more C down the profile than the fava system and 14 Mg C ha^{-1} more than the maize system, corresponding to a 12% to 14% increase in total carbon storage due to incorporation of clover into rotation (Table 4). These managements effects were also indicated by the models of SOC concentration, which revealed highly significant interactive effects of cropping system, fertilization and soil depth on SOC concentration, with SOC concentration enhanced in the upper half of the profile under PK fertilization within the clover system compared with the maize and fava systems. However, the differences between cropping systems were not significant without fertilization nor under full NPK fertilization.

The regression tree analysis indicated that soil C densities under the contrasting cropping systems induced depth specific responses to differing nutrient availability (Fig. 2). The analysis revealed a strong and clear influence of soil depth on C densities with the primary split separating the upper sampling depth (0–30 cm, topsoil) from the other sampling depths (subsoil, 30–100 cm). In the topsoil, C storage was positively associated with plant available K. Moreover, within this depth legumes (primarily clover) enhanced C density in soils with greater plant available K (> 152.4 mg kg^{-1}) but not receiving N fertilizer. This effect was more pronounced with high P availability (Fig. S5). In the 30–50 cm depth, the clover system was enhanced in C compared with fava and maize, but in the deep subsoil (70–100 cm), soil under fava was depleted in C compared with that under clover and maize. Interestingly, C storage at 50–70 cm depth was affected by plant available P, with higher plant available P associated with lower C density.

N densities were highly significantly reduced within the fava system compared with the clover or maize systems (Fig. 3), with effects most pronounced under PK fertilization. In contrast to the predominantly surficial response of SOC to crop rotation under PK fertilization, the differences in soil N density arising from different crop rotation systems were detectable down the entire soil profile. The fava system under PK fertilization stored around 1.8 Mg ha^{-1} less N than either fertilized clover systems. Under PK fertilization, in the upper half of the profile the N density within the maize system reflected that of the fava system but in the lower profile it mirrored the clover system. As a result, it was not statistically distinct from either of the legumes.

In contrast to the interactive management effects on SOC indicated by the models, the models of soil N did not indicate interactive management effects. Despite the high N_{min} inputs in the NPK fertilized plots, there was no significant effect of full mineral fertilization on N storage compared with the plots not receiving N fertilizer (Fig. S6). However, across all fertilization plots, the models indicated significant effects of cropping system on N storage, with the model coefficients indicating significantly lower N storage in the fava system than the clover system.

The regression tree analysis indicated different relationships between cropping system, fertilization and depth on soil N densities than those revealed for C densities (Fig. 4). Here the primary split was

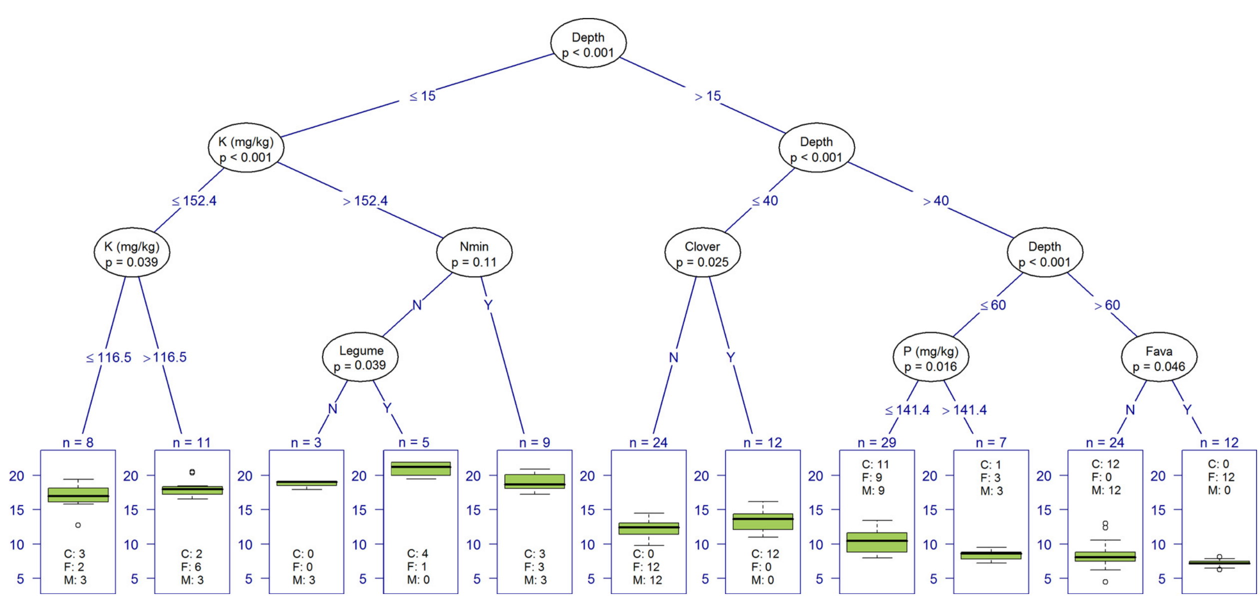

Fig. 2. Regression tree of soil organic carbon density as a function of cropping system, nutrient availability and soil depth. Depth represents the mean sampling depth in cm, Nmin is a factor for the application of mineral N fertilizer, K [mg kg^{-1}] and P [mg kg^{-1}] are the soil concentrations of plant available K and P respectively, Fava and Clover are factor variables for the respective cropping systems. C, F and M in the terminal nodes represent the number of cases of clover, fava and maize cropping system in the respective terminal nodes.

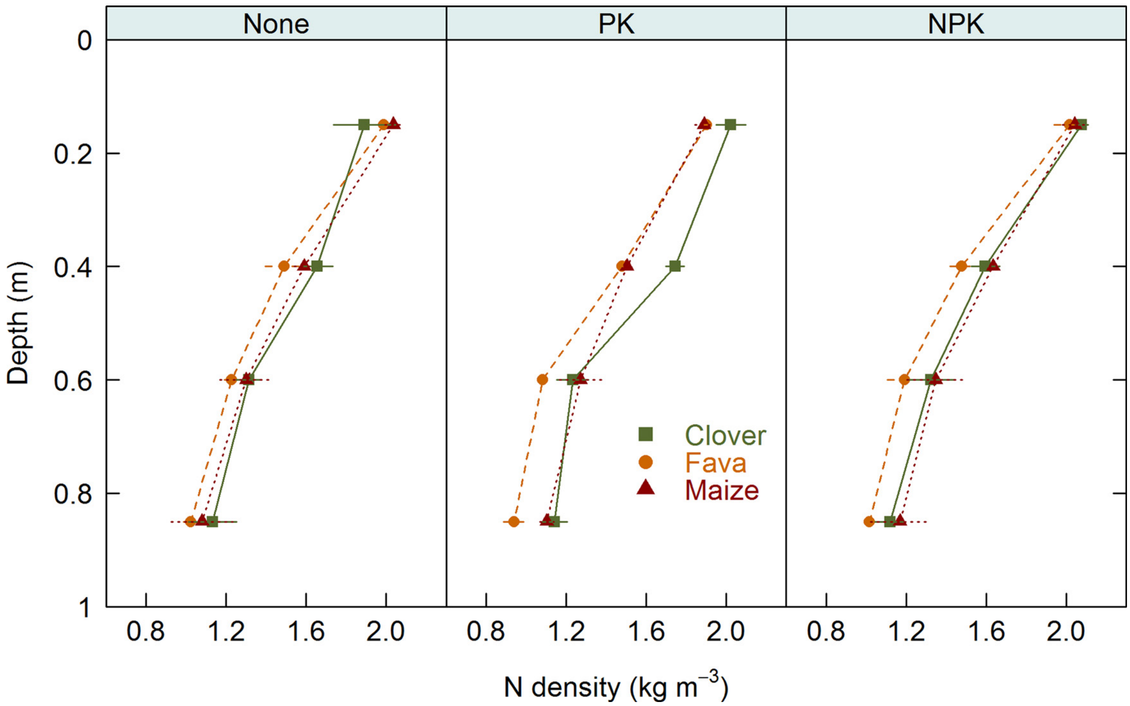

Fig. 3. Soil nitrogen density under different crop rotations systems and fertilization regimes. Dots represent the mean, error bars 1 sd.

between the upper (0–50 cm) and lower (50–100 cm) halves of the soil profile. In the plough layer, N storage was positively associated with plant available K. In the 30–50 cm depth, fava stored significantly less N than clover or maize, and clover likely stored more N than maize. In the 50–70 cm depth, plant available P was negatively associated with N storage. Moreover, for low plant available P contents ($< 48.4\ \mathrm{mg\ kg^{-1}}$) legumes (predominantly fava) were associated with lower N storage, although higher plant available P contents ($> 48.4\ \mathrm{mg\ kg^{-1}}$) under clover led to an enhancement of N. In the deep subsoil, N storage was significantly lower under fava than maize and clover.

The models of the stable C isotopic composition indicated strong effects of cropping system and depth on $\delta^{13}C$ signatures, as well as interactions between cropping system, depth and fertilization affecting $\delta^{13}C$. There was no significant relationship between depth and C isotopic composition for the maize cropping system, but there were highly significant depth relationships for the two legume based systems, which were enhanced in ^{13}C in the deep subsoil (Fig. 5). As such, separate models were evaluated for topsoil and for subsoil below 30 cm.

In the topsoil, all three crop rotation systems were highly significantly isotopically distinct from each other under all fertilizer regimes. Across all fertilization plots, the $\delta^{13}C$ of the maize system was highest, averaging $-25.6 \pm 0.0‰$, or $0.6 \pm 0.2‰$ enriched in ^{13}C compared with the clover system ($-26.1 \pm 0.1‰$), and $1.2 \pm 0.2‰$ enriched compared with the fava system ($-26.8 \pm 0.1‰$). In the

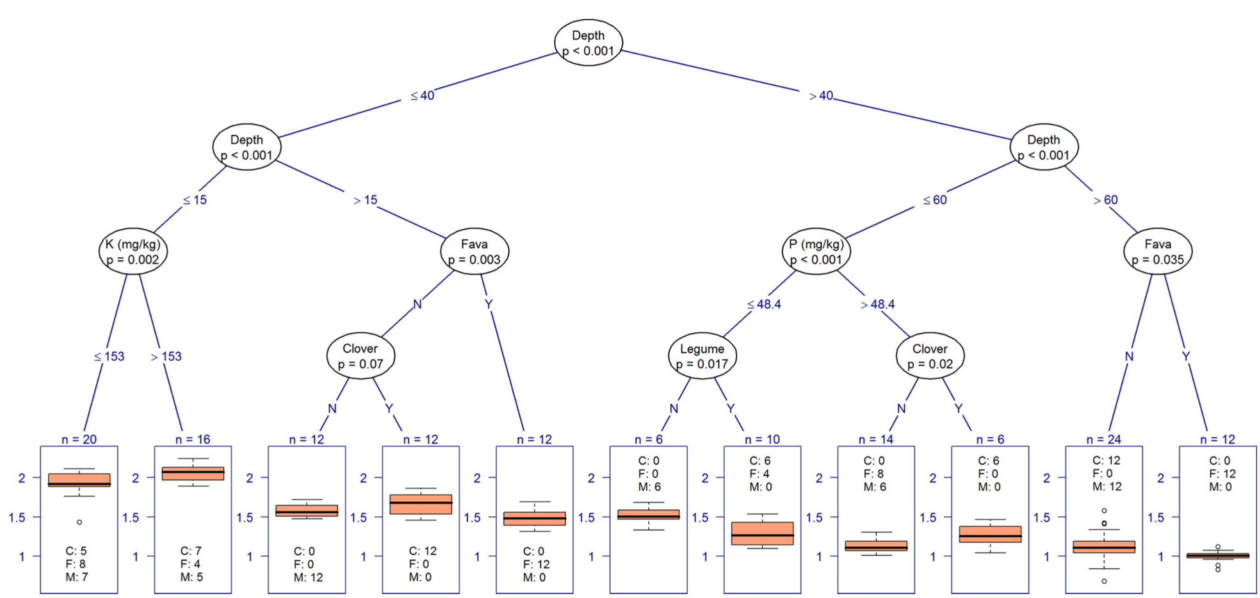

Fig. 4. Regression tree of soil nitrogen density as a function of cropping system, nutrient availability and soil depth. Depth represents the mean sampling depth in cm, K [mg kg^{-1}] and P [mg kg^{-1}] are the soil concentrations of plant available K and P respectively, Fava and Clover are factor variables for the respective cropping systems. Legume is a factor variable for both clover and fava systems. C, F and M in the terminal nodes represent the number of cases of clover, fava and maize cropping system in the respective terminal nodes.

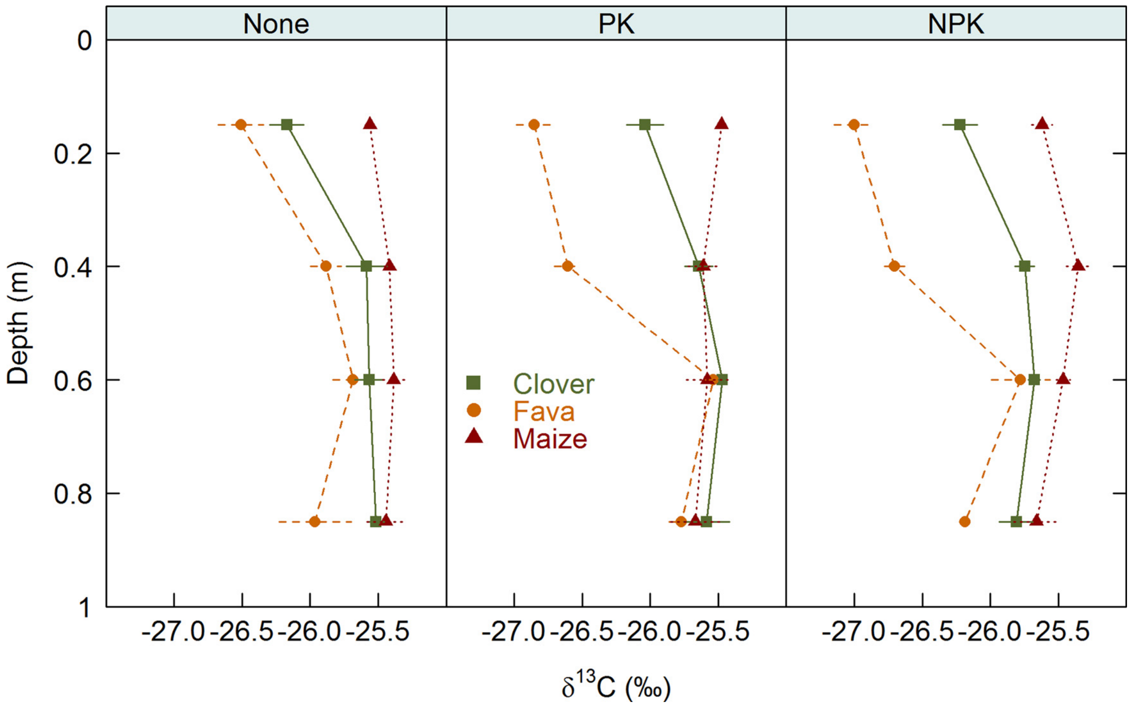

Fig. 5. Stable C isotopic composition under different crop rotations systems and fertilization regimes. Dots represent the mean, error bars 1 sd.

subsoil, the maize system remained highly significantly enriched in ^{13}C compared with both the clover (0.4 ± 0.2‰ difference) and fava (0.5 ± 0.2‰) systems. These differences were not apparent in the PK fertilized system in the lower half of the profile. Fertilization did not significantly affect stable C isotopic composition.

The N-isotopic composition peaked at 30–50 cm, thus lacking the simple depth trend evident within the C isotopic composition (Figs. 5 & 6). As such, separate $\delta^{15}N$ models were fitted to the surface and subsoil data. In the topsoil, both cropping system and fertilization had significant effects on N-isotopic composition, though the models did not indicate interactions between these variables. The topsoil $\delta^{15}N$ of the fava system was 0.5 ± 0.4‰ lower than either the maize or the clover system, which did not differ significantly from each other. NPK fertilization lowered the $\delta^{15}N$ by 0.6 ± 0.4‰ compared with the unfertilized treatments. Although the PK fertilized plots were enriched in $\delta^{15}N$ by 0.4 ± 0.4‰ compared with the NPK fertilized systems, the difference was only significant at $p = 0.07$. There were no statistical differences between the unfertilized and PK fertilized systems. In the subsoil, the fava system was also highly significantly depleted in $\delta^{15}N$, with depletion similar to that at the surface, namely 0.5 ± 0.2‰ lower

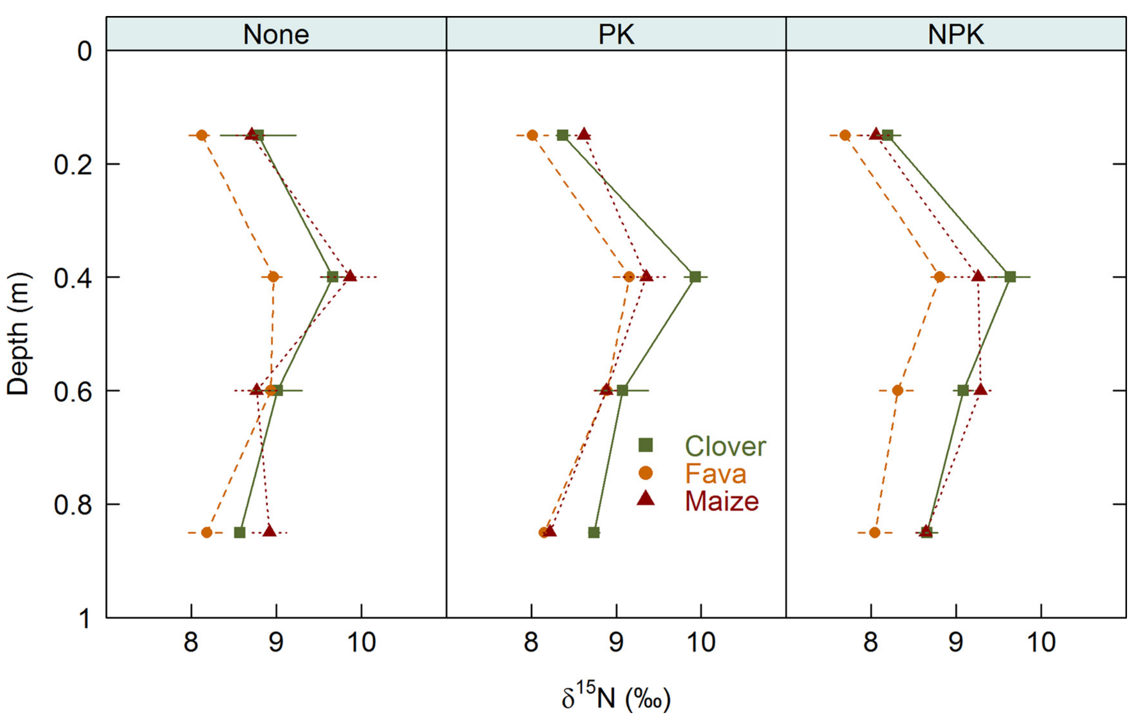

Fig. 6. Stable N isotopic composition under different crop rotations systems and fertilization regimes. Dots represent the mean, error bars 1 sd.

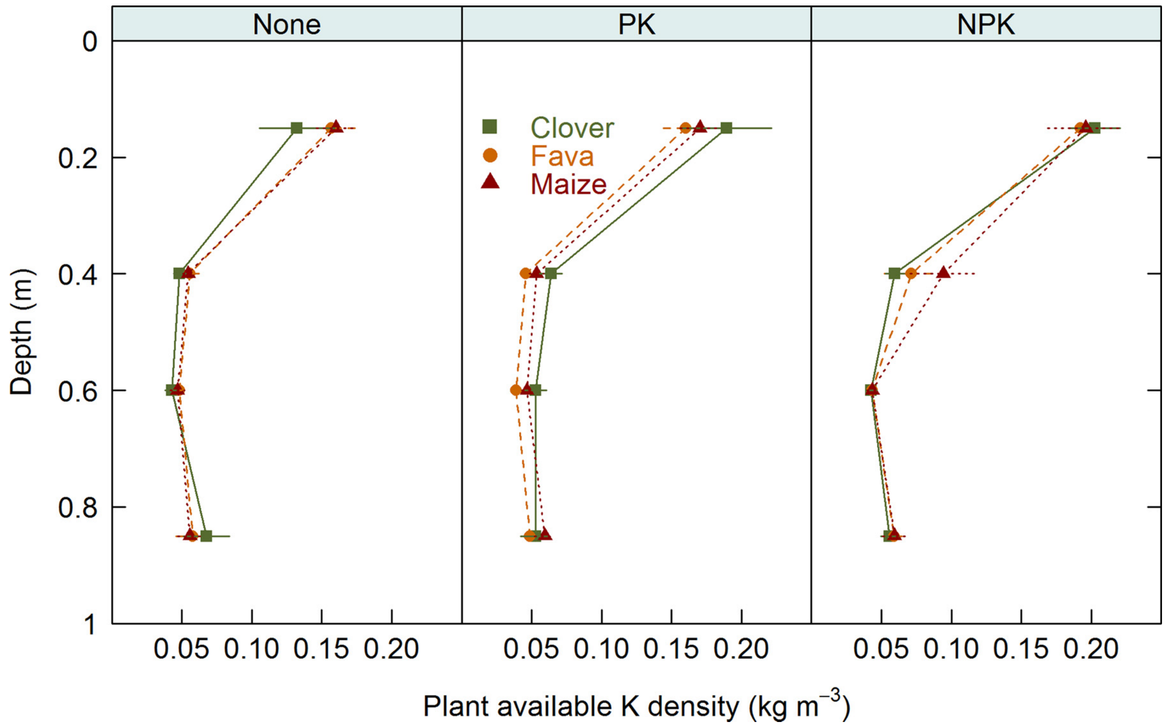

Fig. 7. Plant available soil potassium density under different crop rotations systems and fertilization regimes. Dots represent the mean, error bars 1 sd.

than the clover system and 0.4 ± 0.2‰ lower than the maize system. The shift in N isotopic composition associated with fertilization in the topsoil was not detected in the subsoil.

3.4. Fertilization and crop rotation effects on soil potassium and phosphorus

Due to violation of assumptions in the models of soil K, effects of management on K storage were analysed using separate models for the topsoil and subsoil depths. In the topsoil, fertilization increased plant-available K density significantly from 0.150 ± 0.039 kg m^{-3} without fertilization to 0.197 ± 0.036 kg m^{-3} under NPK fertilization, although PK fertilized plots were not statistically distinct from the other fertilization regimes (Figs. 7 & S8). This trend continued within the 30–50 cm depth, with K storage significantly higher under NPK fertilized plots (0.075 ± 0.036 kg m^{-3}) than unfertilized plots (0.052 ± 0.010 kg m^{-3}). Below 50 cm, neither K concentrations nor K densities were affected by management.

Although P storage mirrored the depth trend of δ^{15}N, with a tendency to a minimum at 30–50 cm, the coefficient of variation was relatively high (Table 3, Figs. 8 & S9) and no significant differences in plant-available P were detected. However, the models indicated a likely increase in plant available P under fava beans ($p = 0.06$) compared with the clover system.

4. Discussion

4.1. Mineral N fertilization and soil physico-chemical properties

One uniting feature of full NPK fertilization on soil properties were the effects on soil acidity and electrical conductivity. As these effects were not observed without additional N_{min}, the causal relationship with mineral N fertilizer is clear. The minor but significant increase in acidity indicates that the addition of lime with mineral N fertilizer is insufficient to counteract the acidification associated with mineral N fertilization, which is permanent and associated with Ca and Mg losses, losses of cation exchange capacity and enhanced exchangeable Al and Fe (Barak et al., 1997; Johnston et al., 1986), though Ca losses are likely compensated for by the lime addition in the calcium ammonium nitrate fertilizer. Interestingly, although the effects were greatest at the surface, the subsoil also revealed higher electrical conductivities and lower pH than plots not receiving N_{min}, strongly suggesting leaching of N_{min} down the profile. However, the current status of these soils indicates pH in a near optimum range, most likely a result of the naturally high fertility of these soils and the addition of lime with N fertilizer, and overall low soil salinity (< 200 µS cm^{-1}).

Intriguingly, NPK fertilization had no substantial impacts on C and N densities. Despite large N_{min} additions, there was little evidence of soil N enhancement within the profile compared with unfertilized systems, consistent with complete compensation of N_{min} addition by plant uptake and system losses. Similarly, the similar (clover and maize) or marginally lower (fava) C densities under NPK fertilization than in the unfertilized plots indicate that although N fertilization enhances aboveground plant productivity (Tab. S1), this does not promote soil organic matter storage. These findings are in contrast to results from long-term field experiments in Sweden (Kätterer et al., 2014). Here various explanations are possible.

Firstly, plants may respond to enhanced high levels of topsoil nutrient availability - namely via N_{min} fertilization - by reducing root growth (Svoboda and Haberle, 2006) and rooting depth (Robinson et al., 1994), which becomes energetically unfavourable when nutrients are not limited. This hypothesis is also supported by the increased bulk densities in the subsoil with NPK fertilization compared with the unfertilized systems, which suggest a reduction in rooting depth due to NPK fertilization under clover and maize rotations. Similarly, the higher bulk densities in the subsoil within the NPK fertilized maize system than the PK fertilized maize system also indicate differential rooting depths as a response to N fertilization. Specifically, our results suggest that the deeper rooting is stimulated by N deficiency and results in N scavenging in the subsoil. These results indicate that crops can respond to N limitation by shifting the ratio of above to belowground biomass, with aboveground biomass responding more positively to N fertilization the root biomass (Mahli and Lemke, 2007). As such, despite greater yields after fertilization, SOC may not be not increased, as was observed in our study.

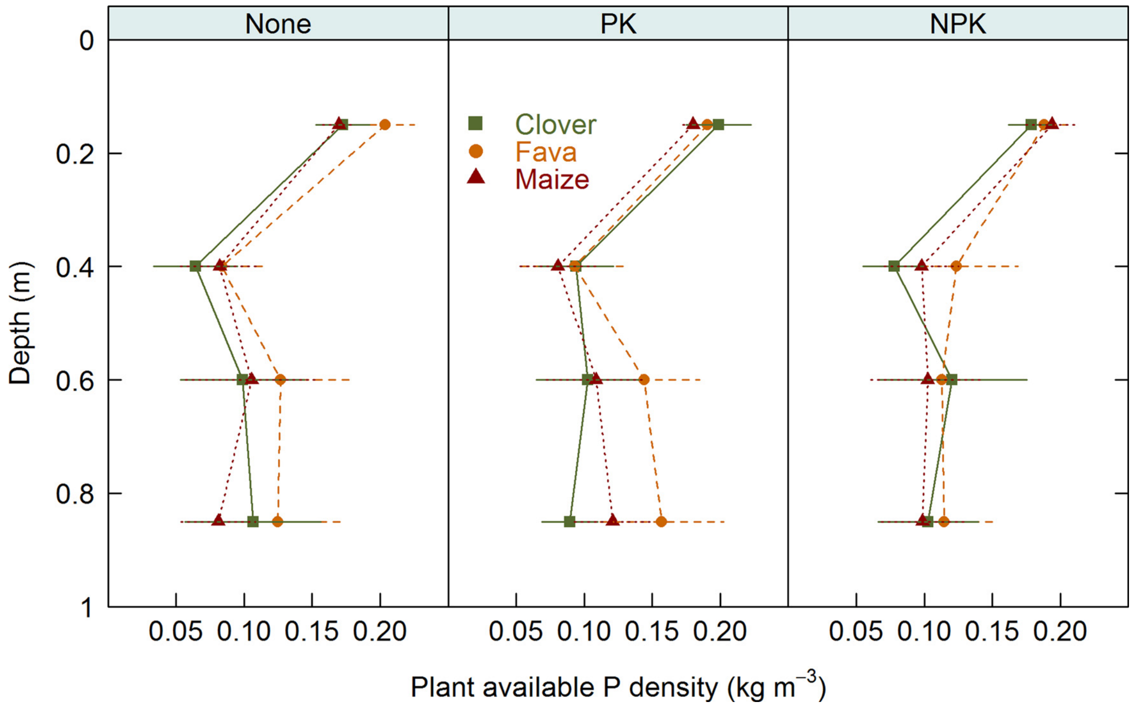

Fig. 8. Plant available soil phosphorus density under different crop rotations systems and fertilization regimes. Dots represent the mean, error bars 1 sd.

Secondly, N fertilization can promote turnover of plant material by microorganisms, increasing decomposition rates (Neff et al., 2002). As such, any additional root biomass produced after fertilization may be compensated for by additional microbial turnover, without a concurrent enhancement in SOC storage. Of course, one process does not preclude the other and the SOC storage is the product of both changes in SOC input and turnover.

Interestingly, subsoil bulk densities were not higher in the fava system after fertilization, suggesting that deeper fava roots are not a response to nutrient limitation. Fava is a 'thirsty' crop which can access water from over 2 m deep in the soil, and root length density in the subsoil increases in response to subsoil water availability (Bryla et al., 2003). As such, it is possible that the deeper rooting of the fava system indicates scavenging for deeply stored water. Indeed, the overall lower $\delta^{13}C$ signatures under fava indicate proportionally greater contributions of plant rather than microbial biomass (Ehleringer et al., 2000) in the subsoil, supporting greater fava rooting depths. Additionally, the enhanced subsoil concentrations of plant available P within the fava system support this hypothesis of enhanced deep rooting, as fava roots mobilize P, especially under additional P fertilization, making it available to subsequent crops (Nuruzzaman et al., 2005).

It is important to note that these differences in rooting depth are not limited to the first year crops, but are likely induced in other crops within the rotation too. Specifically, the C isotopic signatures of the maize system resemble soils dominated by C3 plants, indicating overall low inputs of maize carbon into the soil system. Modelling the shift in C isotopes (Balesdent and Mariotti, 1987) in the maize system compared with the clover and fava systems suggests that in the surface horizon 8 ± 3% of SOC is maize-derived, with only 2 ± 2% of C derived from maize in the deep subsoil. This indicates that maize roots are responsible for significantly less proportional C input, especially into the deep subsoil, than could be expected from their 1 in 4 year annual growth, suggesting that the roots of the cereal crops in the rotation are more important for C storage than the maize crop.

4.2. C and N enhancement due to legumes without mineral N fertilization

Contrasting with the lack of influence of NPK fertilization, PK fertilization induced both differentiated crop and depth dependent effects on soil C and N densities. Within the clover system the C and N density gains in the upper 50 cm of the soil profile suggest that biomass production of clover is limited by either P, K or both, consistent with studies reporting that PK fertilization stimulates clover growth (Wani et al., 1995, Oram et al., 2014), resulting in greater input of biomass and storage of SOC and nitrogen.

Within the topsoil, this response of the clover system to PK fertilization is primarily attributable to increased SOC concentrations, as bulk density was unaffected by cropping system and fertilization, implying higher inputs of biomass in the clover system than the fava and maize systems. The regression tree analysis suggests that K availability is the driver of this higher biomass production and positive SOC storage response. Here it must be stressed that the response is likely to occur in both above and belowground biomass, as the aboveground clover biomass is ploughed into the field as a green mulch. However, at 30–50 cm both the C concentration and the soil bulk density increased under clover compared with maize and fava. This suggests that the additional C storage does not result from enhanced subsoil rooting, which would lower bulk density, but is associated with translocation of C down the profile, e.g. via bioturbation or as dissolved organic carbon. Further, the enhanced subsoil bulk densities suggest that the promotion of clover biomass production due to PK fertilization provided a source of N for subsequent crops, which responded with reduced rooting depths.

That the additional C sequestration and N storage in the PK fertilized clover extends below the plough horizon into the subsoil highlights the potential to manage soils for subsurface SOC sequestration (Hobley and Wilson, 2016). Using crimson clover as a green mulch in combination with P and K fertilization therefore has the potential to sequester soil carbon. Over the 34 years of the experiment, the gains in whole profile SOC in the PK fertilized clover system represent an average annual gain of 4.1‰ compared with the maize system and are therefore well aligned with the goals of the 4 per 1000 initiative arising from the Lima-Paris Action Agenda (Chabbi et al., 2017).

As conventional management, i.e. mineral fertilization, ploughing and production of cash crops, is known to deplete SOC down the whole profile (Hobley et al., 2017; Hobley and Wilson, 2016; Shahbaz et al., 2017; Wiesmeier et al., 2015), it is possible that this difference in C storage does not truly represent a C gain but is merely a result of reduced losses in the clover system compared with the maize system. To test whether the differences in C storage represent a gain in C or are potentially the result of a smaller C loss, the C storage in the maize NPK fertilization system was compared with a long-term trial established in 1954 at the same site with conventional agricultural management (fertilization and crop rotation, University of Gießen) and sampled during the same sampling campaign. There was no significant difference in C storage down the profile ($p > 0.1$), between the site under management for ~30 years and that under management for ~60 years, indicating that the long-term management of the soil has induced an apparent steady state. As such, we conclude that a PK fertilized clover system would represent a true SOC gain for similar systems which are converted from long-term conventional management to clover mulch rotation. We highlight that the 4.1‰ gain was detected down a full metre of soil, as opposed to the traditional view of 'reactive' topsoil in many soil investigations (e.g. Blanco-Canqui et al., 2017). Only sampling the traditional topsoil (0–30 cm. e.g. Hobley et al., 2015) accounting depth is therefore insufficient in systems where subsoil organic carbon enhancement occurs, such as in our study.

Interestingly, topsoil SOC density was statistically indistinct in the unfertilized fava system from the PK fertilized clover system, indicating surface carbon sequestration potential without additional fertilizer in unfertilized fava plots, with the regression tree indicating that K availability is decisive in enhancing C storage. However, the contradictory results of the linear models and regression tree analysis show that results for the fava system are not conclusive.

On the other hand, the lack of positive response of SOC storage within the maize system implies that PK is not limiting to maize growth at this site, likely due to N being the primary limiting factor for growth. It is also possible that PK fertilization may have shifted the decomposition dynamics under the maize rotation, enhancing SOM turnover and therefore leading to no net SOC gain. However, the isotopic signatures and bulk densities indicate that this is not the case, as there was scant evidence of input and incorporation of maize C into and root growth within the profile, with input limited to the surface horizon. As such, N limitation to SOC production seems to be the greater influence on SOC dynamics in the maize system with PK fertilization.

Overall, the different responses in SOC storage under the fava and clover systems contradict our hypothesis that legumes boost SOC storage, which was only clearly supported for clover under PK fertilization. However, the hypothesis that SOC response to management is depth dependent was confirmed. These differential depth, crop and nutrient responses help explain why some soils sequester SOC under legumes whereas others do not (Robertson et al., 2015).

4.3. C and N dynamics are not coupled down the profile but occur in distinct hotspots

Intriguingly, both the linear models and the regression tree analyses suggest that the processes governing top- and subsoil C and N dynamics are not entirely coupled, particularly in the subsoil. The enrichment of ^{15}N at the soil depth of 30–50 cm indicates that this depth is a hotspot for N processing (van Groenigen et al., 2015), potentially attributable to loss of isotopically light gaseous N during nitrification and denitrification processes, which lead to strong isotopic fractionation and enrichment of ^{15}N in residual soil N pools (Kriszan et al., 2014; Robinson, 2001). As denitrification is an anaerobic process, it more likely to occur in the subsoil, whereas the greatest C turnover occurs aerobically, helping to explain differential zones of N and C turnover and the decoupling of these processes.

Alternatively, the enrichment in ^{15}N at this depth could also arise from N fractionation during plant assimilation, which leads to a less pronounced but significant enrichment of ^{15}N in the soil compared with gaseous transformation (Robinson, 2001). This is supported by the largest deviation from the topsoil in P storage at 30–50 cm, suggesting that this region of the subsoil might develop into a key reservoir of soil nutrient cycling after prolonged duration of cropping (Amelung et al., 2015). Below 50 cm, the similarity of $\delta^{15}N$ signatures to those in the topsoil indicate that this zone is not a hotspot for N turnover and uptake.

In contrast with the N isotope results, the C isotopes only indicate a clear shift towards more microbially processed C (Ehleringer et al., 2000, Wynn et al., 2004) below 50 cm, whereas the 30–50 cm depth appears to be (at most) a transitional zone between surface and subsoil processes. This suggests that the N hotspot at 30–50 cm depth is not solely attributable to microbial N processing but that plant uptake influences N cycling at this depth. This helps to explain the decoupling of N and C below the topsoil: while plant roots introduce C into the subsoil they also scavenge for N in the subsoil (Haberle et al., 2006), which is then harvested from the system, decoupling these cycles.

Interestingly, the subsoil C isotopes within the fava system do not exhibit the strong shift to isotopically enriched, microbial C (Blair et al., 1985) shown by the clover and maize systems, providing evidence for greater proportional inputs of root C. Moreover, the lower N storage in the subsoil under fava could also be explicable by fava deep rooting to fulfil water demand, altering the water budget and therefore lowering soil N translocation to and concentrations in the subsoil. Additionally, uptake of subsoil N by deep fava roots is likely, as legumes are often incapable of fulfilling all their N needs via fixation, despite favourable N budgets compared with non-leguminous systems (Wani et al., 1995). Once again, we note that subsoil N scavenging probably occurs by subsequent crops such as wheat, which can utilize N from deep within the soil to depths of greater than a metre (Haberle et al., 2006), resulting in overall lower N storage in the deep subsoil within the fava system.

Lastly, although we used multiple proxies, namely C and N densities and concentrations, isotopes and available nutrients to infer whole profile C and N dynamics, future studies into direct drivers of subsoil C and N processing, specifically plant rooting and microbial activity, are required to elucidate subsoil C and N cycles. Specifically, our isotopic data suggest that there are subsoil hotspots for N processing, which are distinct from hotspots for C processing. Taken together our results strongly suggest that subsoils are dynamic (Fontaine et al., 2007; Hobley et al., 2017), with cropping systems actively responding to management and nutrient availability, highlighting the possibility for management optimization to target site specific issues such as C depletion or N leaching.

4.4. Outlook

Long-term mineral N fertilization did not promote SOC storage within the profile, indicating decoupling of above and belowground biomass production due to mineral N fertilization. In contrast, the long-term response of the studied soils with regards to carbon and nitrogen storage was differentiated according to cropping system, management and nutrient availability:

- C inputs from maize are small and negligible in the subsoil;
- Fava beans, in turn, appeared to reduce N translocation probably via deeper rooting systems, so that they may be an attractive option for managing sites susceptible to N leaching;
- Clover as a green mulch under PK fertilization is most promising for carbon sequestration at this site. Over 30 years of green mulching added an average of 4.1‰ C per annum to the soil profile compared with conventional crop rotation and fertilization management.

This C enhancement occurred below the plough horizon, highlighting the potential for subsurface C sequestration via optimized crop and fertilization management, even in sites of naturally high fertility. Assessment of C and N storage in agricultural soils should not be limited to topsoil horizons but must include cycling in the deeper profile.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.agee.2018.06.021.

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